Assume/Guarantee Contract of `GyroChoose`

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1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet : StateFlag, JoinTotal′, gyroStatus0′, SignFlag′, clock, clock′

Inputs :

- `StateFlag` $\mapsto$ StateFlag : $\mathbb{N} \rightharpoonup \mathbb{R}$

Outputs :

- `JoinTotal` $\mapsto$ JoinTotal : $\mathbb{N}$
- `gyroStatus0` $\mapsto$ gyroStatus0 : $\mathbb{N}$
- `SignFlag` $\mapsto$ SignFlag : $\mathbb{N} \rightharpoonup \mathbb{R}$

In-Outputs : None

States : None

Parameters : None

Clocks : clock, clock′ : $\mathbb{R}_{\geq 0}$

2 AG Contract

Assumptions :

$$Assumptions \triangleq \mathsf{len}(\mathsf{StateFlag}) \geq 9$$

Guarantees :

$$\begin{aligned}
Guarantees &\triangleq Check_1 \wedge Check_2 \wedge Check_3 \\
Check_1 &\triangleq \mathsf{len}(\mathsf{SignFlag}') = 9 \\
&\quad \wedge \mathsf{clock}' = \mathsf{clock} \\
Check_2 &\triangleq \forall \mathsf{JoinTotal}' \leq i < 9 \cdot \mathsf{SignFlag}'[i] = 0 \\
Check_3 &\triangleq \exists \mathsf{tempFlag} : \mathbb{N} \rightharpoonup \mathbb{R} \cdot \big(\mathsf{len}(\mathsf{tempFlag}) = 9 \wedge Check_4 \wedge Check_5 \wedge Check_6 \wedge Check_7 \big) \\
Check_4 &\triangleq \forall 0 \leq i < 9 \cdot \big(\mathsf{tempFlag}[i] = 1 \lhd \mathsf{StateFlag}[i] = \texttt{0xEB} \rhd \mathsf{tempFlag}[i] = 0 \big) \\
Check_5 &\triangleq \mathsf{JoinTotal}' = \textstyle\sum_{j=0}^{8} \mathsf{tempFlag}[j] \\
Check_6 &\triangleq \forall 0 \leq i < \mathsf{JoinTotal}' \cdot \mathsf{SignFlag}'[i] = \min\{k \mid \textstyle\sum_{j=0}^{k} \mathsf{tempFlag}[j] = i\} \\
Check_7 &\triangleq \mathsf{gyroStatus0}' = \textstyle\sum_{i=0}^{8} \mathsf{tempFlag}[i] \times 2^i
\end{aligned}$$